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Module - V
Statistical Inference -

Sampling variables, central limit theorem and confidence limit for unknown mean. Test of significance for means of two small samples, Student's 't' distribution, chi-square distribution as a test of goodness of fit.

Statistics :-

Any function of the sample value is known as a statistics.

Ex:- sample mean, sample median, sample variance etc.

Sampling Distribution:-

A sampling distribution is a distribution of a statistic over all possible samples. That is sampling distribution is the probability distribution of the statistics.

OR

generated frequency distributions of a samples is known as sampling distribution.

Sampling variables:-

Variables sampling is the process used to predict the value of a specific variable within a population ex:- a limited sample size can be used to compute the average accounts receivable balance, as well as a statistical derivation of the plus @ minus range of the total receivables.

The central Limit Theorem:-

The central limit theorem states that the sample mean $\bar{x}$ follows approximately the normal distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$ (is also called standard error),

ie $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

where μ , σ are mean and standard deviation of the population from where the sample was selected and the sample size becomes large ($n \geq 30$).

Degrees of Freedom:-

Degrees of freedom refer to the maximum number of logically independent values, which may vary in a data sample.

Degrees of freedom are calculated by subtracting one from the number of items within data sample ($n-1$).

Description	Population notation	Sample Notation
Size	N	n
Mean	μ	$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$
Variance	σ^2	$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$
Standard deviation	σ	$s = \sqrt{\frac{1}{n-1} \sum (x - \bar{x})^2}$

Confidence Intervals:-

From the table of normal distribution, we find that 95% of the area lies between $z = -1.96$ and $z = +1.96$. In other words we can say with 95% confidence that z lies between -1.96 and $+1.96$. Further 5% level of significance is denoted by $z_{0.05}$.

Thus we can write the verbal statement in the mathematical form,

$$\bar{x} - 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) \quad \text{--- (1)}$$

$$\bar{x} - 2.58 \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + 2.58 \left(\frac{\sigma}{\sqrt{n}} \right) \quad \text{--- (2)}$$

Thus we say that (1) is the 95% confidence interval and (2) is the 99% confidence interval.

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Confidence Interval C.I = Mean $\pm$ $z \left(\frac{\text{Standard deviation}}{\sqrt{\text{Sample Size}}} \right)$

$$= \mu \pm z \frac{\sigma}{\sqrt{n}} \quad (\text{or}) \quad \bar{x} = \mu \pm z \frac{\sigma}{\sqrt{n}}$$

confidence level	99%	95%
z	2.58	1.96

Problems

1. State central limit theorem. Use the theorem to evaluate $P[50 < \bar{X} < 56]$ where $\bar{X}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$.

Soln:- The central limit theorem states that the sample mean $\bar{X}$ follows approximately the normal distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$

[It is also called standard error]

ie $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$, where μ, σ are mean and standard deviation of the population from where the sample.

→ Given,

Sample size $n = 100$

Mean of the population $\mu = 53$

Variance of the population $\sigma^2 = 400 \Rightarrow \sigma = \sqrt{400} = 20$

$$\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$$

$$\Rightarrow \bar{X} \sim N(53, \frac{20}{\sqrt{100}})$$

$$\Rightarrow \bar{X} \sim N(53, 2)$$

$$\therefore \text{WKT } Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$\Rightarrow Z = \frac{\bar{X} - 53}{\frac{20}{\sqrt{100}}} = \frac{\bar{X} - 53}{2}$$

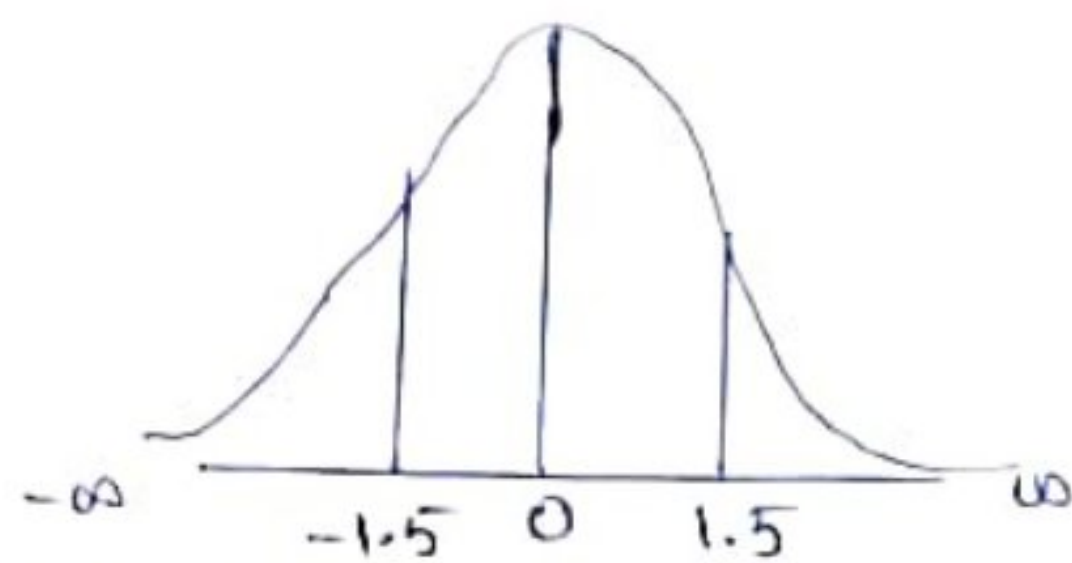
$$\therefore \text{At } \bar{X} = 50 \Rightarrow Z = \frac{50 - 53}{2} = \frac{-3}{2} = -1.5 = Z_1$$

$$\text{At } \bar{X} = 56 \Rightarrow Z = \frac{56 - 53}{2} = \frac{3}{2} = 1.5 = Z_2$$

$$\therefore P(50 < \bar{X} < 56) = P(-1.5 < Z < 1.5)$$

$$= 2P(0 < Z < 1.5) = 2P(1.5) = 2(0.4332)$$

$$\therefore P(50 < \bar{X} < 56) = \underline{0.8664}$$



Standard area represents probability $P(50 < \bar{X} < 56)$

2. An unknown distribution has a mean of 90 and a standard deviation of 15. Samples of size $n=25$ are drawn randomly from the population. Find the probability that the sample mean is between 85 and 92.

Soln:- Given,

Sample size $n=25$

Mean of the population $\mu=90$

Variance of the population $\sigma=15$

$$\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$$

$$\Rightarrow \bar{X} \sim N(90, \frac{15}{\sqrt{25}})$$

$$\Rightarrow \bar{X} \sim N(90, 3)$$

$$\therefore \text{w.k.T } z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$\Rightarrow z = \frac{\bar{X} - 90}{15/\sqrt{25}}$$

$$\Rightarrow z = \frac{\bar{X} - 90}{3}$$

$$\therefore \text{At } \bar{X} = 85 \Rightarrow z = \frac{85 - 90}{3} = \frac{-5}{3} = -1.66$$

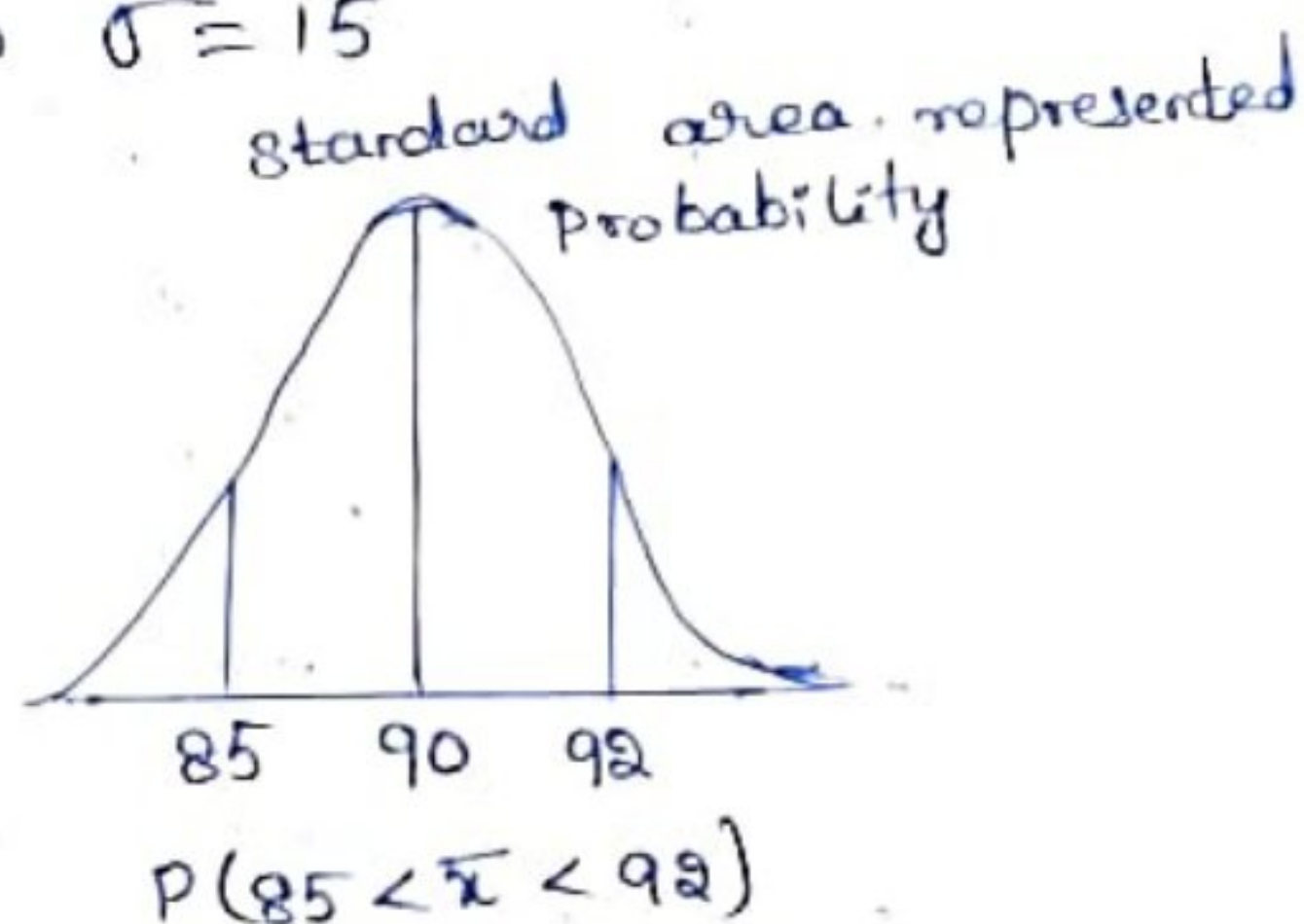
$$\therefore \text{At } \bar{X} = 92 \Rightarrow z = \frac{92 - 90}{3} = \frac{2}{3} = 0.66$$

$$\therefore P(85 < \bar{X} < 92) = P(-1.66 < z < 0.66)$$

$$\Rightarrow P(-1.66 < z < 0.66) = P(0 < z < 1.66) + P(0 < z < 0.66)$$

$$= 0.4515 + 0.2454$$

$$\Rightarrow P(-1.66 < z < 0.66) = 0.6969$$



3. A random sample of size 64 is taken from an infinite population having mean 112 and variance 144. Using central limit theorem, find the probability of getting the sample mean $\bar{X}$ greater than 114.5

$\Rightarrow$ Soln:-

Given,

Sample size $n=64$

Mean of the population $\mu=112$

variance of the population $=\sigma^2=144 \Rightarrow \sigma=12$

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$$\Rightarrow \bar{X} \sim N\left(112, \frac{12}{\sqrt{64}}\right)$$

$$\Rightarrow \bar{X} \sim N(112, 1.5)$$

$$\therefore \text{wkt } z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$\Rightarrow z = \frac{\bar{X} - 112}{12/\sqrt{64}}$$

$$\Rightarrow z = \frac{\bar{X} - 112}{1.5}$$

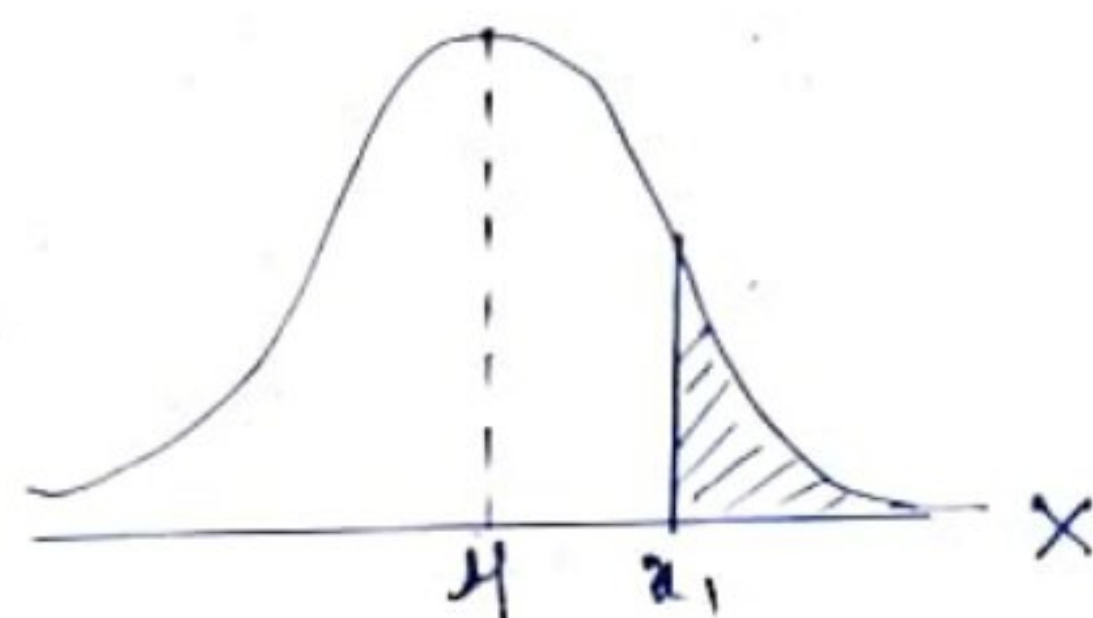
$$\therefore \text{At } \bar{X} = 114.5 \Rightarrow z = \frac{114.5 - 112}{1.5} = 1.66$$

$$\therefore P(\bar{X} > 114.5) = P(z > 1.66)$$

$$\Rightarrow P(z > 1.66) = 0.5 - P(0 < z < 1.66)$$

$$= 0.5 - 0.4515$$

$$\Rightarrow P(z > 1.66) = 0.0485$$



Shaded area represents probability $P(X \geq z_1)$

4. The heights of a random sample of 50 college students showed a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Construct a 99% confidence interval for the mean height of all college students.

→ Soln:-

Given the sample size $n = 50$
 Average height of students (Mean) $\mu = 174.5$ cm
 Standard deviation of the students $\sigma = 6.9$ cm

wkt

confidence level of 99%, the corresponding z value is 2.576.

This is determined from the normal distribution table.

confidence interval $C.I = \text{Mean} \pm z \left(\frac{\text{Standard deviation}}{\sqrt{\text{Sample Size}}} \right)$

$$= \mu \pm z \frac{\sigma}{\sqrt{n}}$$

$$\therefore C.I = 174.5 \pm \left(2.576 \times \frac{6.9}{\sqrt{50}} \right)$$

$$\Rightarrow C.I = 174.5 \pm (2.576 \times 0.9758)$$

$$\Rightarrow C.I = 174.5 \pm (2.5136)$$

The lower end of the confidence interval is =

$$174.5 - 2.5136 = \underline{171.9864}$$

The upper end of the confidence interval is =

$$174.5 + 2.5136 = 177.0136$$

$\therefore$ with 99% confidence interval, the mean height of all college students is b/w 171.9864 centimeters and 177.0136 centimeters.

5. A random sample of size 25 from a normal distribution ($\sigma^2=4$) yields, sample mean $\bar{X}=78.3$. obtain a 99% confidence interval for μ .

→ Soln:-

Given the sample size $n=25$

Mean of sample $\bar{X}=78.3$

Standard deviation $\sigma=2$

we know, confidence level of 99%, the corresponding Z value is 2.58.

This is determined from the normal distribution table.

confidence interval C.I = $\mu = \text{Mean} \pm z \left(\frac{\text{standard deviation}}{\sqrt{\text{Sample Size}}} \right)$

$$= \bar{X} \pm z \frac{\sigma}{\sqrt{n}}$$

$$\therefore \text{C.I.} = \mu = 78.3 \pm \left(2.58 \times \frac{2}{\sqrt{25}} \right)$$

$$\Rightarrow \mu = 78.3 \pm 1.032$$

$$\Rightarrow \text{C.I.} \Rightarrow (78.3 - 1.032, 78.3 + 1.032)$$

$$\text{C.I.} \Rightarrow \underline{(77.268, 79.332)}$$

6. Suppose that 10, 12, 16, 19 is a sample taken from a normal population with variance 6.25. Find at 95% confidence interval for the population mean.

Soln:- Given samples are 10, 12, 16 and 19

$\therefore$ Sample size $n=4$

$$\text{Mean } \bar{X} = 14.25$$

$$\text{variance } \sigma^2 = 6.25$$

$$\Rightarrow \sigma = \sqrt{6.25} = 2.5$$

we know, confidence level of 95%, the corresponding Z value is 1.96, This is determined from the normal distribution table.

$$\text{confidence interval C.I} = \mu = \text{Mean} \pm z \left(\frac{\text{Standard deviation}}{\sqrt{\text{sample size}}} \right)$$

$$= \bar{X} \pm z \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$\therefore \text{C.I.} = \mu = 14.25 \pm \left(1.96 \times \frac{2.5}{\sqrt{4}} \right)$$

$$\Rightarrow \mu = 14.25 \pm 2.45$$

$$\Rightarrow \text{C.I} = (14.25 - 2.45, 14.25 + 2.45)$$

$$\Rightarrow \underline{\underline{\text{C.I} = (11.80, 16.70)}}$$

7 An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed with mean 800 hours and a standard deviation of 40 hours. Find the probability that a random sample of 16 bulbs will have an average life of less than 775 hours.

→ Total number of bulbs $n=16$

An average life of bulbs $\Rightarrow \mu=800$

standard deviation of the bulbs $\Rightarrow \sigma=40$

$\therefore$ we know that $\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

$$\Rightarrow \bar{x} \sim N\left(800, \frac{40}{\sqrt{16}}\right)$$

$$\Rightarrow \bar{x} \sim N(800, 10)$$

$$\therefore Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{x} - 800}{10}$$

$$\text{at } \bar{x}=775 \Rightarrow Z = \frac{775 - 800}{10} = \frac{-25}{10} = -2.5$$

$$\therefore P(\bar{x} < 775) = P(Z < -2.5)$$

$$\Rightarrow P(Z < -2.5) = P(Z > 2.5)$$

$$\Rightarrow P(Z < -2.5) = 0.5 - P(0 < Z < 2.5)$$

$$= 0.5 - \Phi(2.5)$$

$$= 0.5 - 0.4938$$

$$\Rightarrow P(Z < -2.5) = 0.0062$$

Students t - distribution

Let μ be the mean of population, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ be the mean and $s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$ be the standard deviation of a sample, then the student's t - distribution is defined as

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{\bar{x} - \mu\sqrt{n}}{s}$$

Another formula for t - test of two samples is

$$t = \frac{(\bar{x}_2 - \bar{x}_1)}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{where, } s^2 = \frac{n_1 \sigma_1^2 + n_2 \sigma_2^2}{n_1 + n_2 - 2} \quad (\text{or})$$

$$s = \sqrt{\frac{1}{n_1 + n_2 - 2} \left[\sum_{i=1}^{n_1} (x_i - \bar{x}_1)^2 + \sum_{i=1}^{n_2} (x_i - \bar{x}_2)^2 \right]}$$

Problems

1. Find the student's 't' for the following variable volume in a sample of eight values -4, -2, -2, 0, 2, 2, 3, 3 taking the mean of the universe to be zero.

→ Given $\mu = 0$, $n = 8$

$$t = \frac{\bar{x} - \mu\sqrt{n}}{s}$$

$$\bar{x} = \frac{\sum x}{n} = \frac{-4 - 2 - 2 + 0 + 2 + 2 + 3 + 3}{8} = \frac{1}{4} = 0.25$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$= \frac{1}{8-1} [(-4.25)^2 + (-2.25)^2 + (-2.25)^2 + (-0.25)^2 + (1.75)^2 + (1.75)^2 + (2.75)^2 + (2.75)^2]$$

$$= \frac{1}{7} (49.5)$$

$$S^2 = 7.0714$$

$$S = \sqrt{7.0714} = 2.6592$$

$$\therefore t = \left(\frac{0.25 - 0}{2.6592} \right) \sqrt{8} = 0.2659$$

$$t = 0.2659$$

2. Ten individuals are chosen at random from a population and their height in inches are found to be 63, 63, 66, 67, 68, 69, 70, 70, 71, 71. Test the hypothesis that the mean height of the universe is 66 inches [$t_{0.05} = 2.262$ for 9 d.f].

→ Given: $\mu = 66$, $n = 10$

$$\bar{x} = \frac{\sum x}{n} = \frac{678}{10} = 67.8$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S^2 = \frac{1}{10-1} [(63-67.8)^2 + (63-67.8)^2 + \dots + (71-67.8)^2]$$

$$S^2 = 9.07$$

$$S = 3.01$$

$$t = \frac{\bar{x} - \mu}{s} \sqrt{n}$$

$$t = \frac{(67.8 - 66)}{3.011} \sqrt{10} = 1.89 < 2.262$$

Thus hypothesis is accepted at 5% level of significance.

3. A machine is expected to produce nails of length 3 inches. A random sample of 25 nails gave a average length of 3.1 inch with S.D 0.3 Can it be said that the machine is producing nails as per signification? ($t_{0.05} = 2.064$ for 24 degree of freedom)

→ Given: $\mu = 3$, $\bar{x} = 3.1$, $n = 25$, $s = 0.3$

$$t = \left(\frac{\bar{x} - \mu}{s} \right) \sqrt{n}$$

$$t = \frac{0.1}{0.3} \sqrt{25}$$

$$t = 1.67 < 2.064$$

Thus the hypothesis that the machine is producing nails as per specification is accepted at 5% level of significance.

4. A Certain stimulus administered to each of the 12 patients 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. Can it be concluded that the stimulus will increase the blood pressures? ($t_{0.05} = 2.201$ for 11 degree of freedom)

→ Given : $n = 12$

$$\bar{x} = \frac{\sum x}{n} = \frac{31}{12} = 2.583$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S^2 = \frac{1}{11} [(5-2.583)^2 + (2-2.583)^2 + (8-2.583)^2 + (-1-2.583)^2 + (3-2.583)^2 + (0-2.583)^2 + (6-2.583)^2 + (-2-2.583)^2 + (1-2.583)^2 + (5-2.583)^2 + (0-2.583)^2 + (4-2.583)^2]$$

$$S^2 = 9.538$$

$$S = \sqrt{9.538} = 3.088$$

$$\therefore t = \left(\frac{\bar{x} - \mu}{S} \right) \sqrt{n}$$

We take $\mu = 0$ ($\therefore$ stimulus administered does not accompanied with increase in Blood pressure)

$$t = \frac{2.583 - 0}{3.088} \sqrt{12}$$

$$t = 2.8979 \approx 2.9 > 2.201$$

Hence the hypothesis is rejected at 5% level of Significance.

$\therefore$ we conclude with 95% Confidence that the stimulus in general is accompanied with increase in blood pressure.

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5. A random sample of 10 boys had the following

I.Q : 70, 120, 110, 101, 88, 83, 95, 98, 107, 100.

Does this data support the assumption of a population mean I.Q of 100 at 5% level of significance?

(given $t_{0.05} = 2.262$ for 9 degree of freedom).

→ Given the I.Q of 10 boys

x : 70, 120, 110, 101, 88, 83, 95, 98, 107, 100

$$\therefore \bar{x} = \frac{1}{n} \sum x = \frac{972}{10} = 97.2$$

$$\text{Variance, } S^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$$

$$\Rightarrow S^2 = \frac{1}{9} \times 1833.6$$

$$S^2 = 203.73333$$

$$S = 14.2735$$

Given the mean of population $\mu = 100$

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \quad (\text{or}) \quad \frac{(\bar{x} - \mu)\sqrt{n}}{S}$$

$$t = \frac{97.2 - 100}{\left(\frac{14.2735}{\sqrt{10}}\right)} = -0.6203 < 2.262.$$

6. The nine items of a sample have the following values:

45, 47, 50, 52, 48, 47, 49, 53, 51. The sample is drawn from a population whose mean is 47.5. find whether the sample mean differ significantly from the population mean at 5% significance level ($t_{0.05} = 2.31$ for 8 degree of freedom)

Given: $\mu = 47.5$, $n = 9$

$$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$$

$$\bar{x} = \frac{\sum x}{n} = \frac{442}{9} = 49.11$$

$$s^2 = 6.8611 \Rightarrow \boxed{s = 2.6194}$$

$$t = \frac{\bar{x} - \mu}{s} \sqrt{n}$$

$$= \frac{49.11 - 47.5}{2.6194} \sqrt{9}$$

$$t = 1.8439 < 2.31$$

Hence the hypothesis is accepted at 5% level of significance.

$\therefore$ The difference b/w the sample mean and the true mean is not significant.

Test of significance of difference between sample means

Let $(\bar{x}, s_x^2)$ & $(\bar{y}, s_y^2)$ respectively be the mean and variance of the two samples.

Let μ be the population mean and σ^2 be the population variance. we need to test the hypothesis whether the difference between the sample means is significant.

we compute

$$t = \frac{\bar{x} - \bar{y}}{S \cdot \sqrt{1/n_1 + 1/n_2}}$$

where

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

or

$$S^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2 \right]$$

and degrees of freedom $\nu = n_1 + n_2 - 2$

- if $|t| > t_{0.05}$ the difference between the sample means is said to be significant at 5% level of significance.
u'g for 1% level of significance also.

Confidence limits for the population mean μ

we have 95% confidence limits for μ given by $\bar{x} \pm \frac{S}{\sqrt{n}} t_{0.05}$

u'g for 99% confidence limits for μ , $\bar{x} \pm \frac{S}{\sqrt{n}} t_{0.01}$

1] A group of boys and girls were given an intelligence test. The mean score, S.D score and numbers in each group are as follows.

	Boys	Girls
mean	74	70
S.D	8	10
n	12	10

Is the difference between the means of the two groups significant at 5% level of significance ($t_{0.05} = 2.086$ for 20 d.f.)

Solⁿ: - we have by data $\bar{x} = 74$, $s_1 = 8$, $n_1 = 12$ [Boys]

$\bar{y} = 70$, $s_2 = 10$, $n_2 = 10$ [Girls]

we have
$$t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where
$$s^2 = \frac{1}{n_1 + n_2 - 2} \left\{ \sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2 \right\}$$

or
$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

Now
$$s^2 = \frac{12(64) + 10(100)}{20} = \frac{1768}{20} = 88.4$$

$\therefore s = 9.402 \approx 9.4$

hence
$$t = \frac{74 - 70}{9.4 \sqrt{\frac{1}{12} + \frac{1}{10}}} = 0.994$$

$$t = 0.994 < t_{0.05} = 2.086$$

Thus the hypothesis that there is a difference between the means of the two groups is accepted at 5% level of significance.

2] Two types of batteries are tested for their length of life and the following result were obtained.

Battery A : $n_1 = 10$, $\bar{x}_1 = 500$ hrs, $\sigma_1^2 = 100$

Battery B : $n_2 = 10$, $\bar{x}_2 = 500$ hrs, $\sigma_2^2 = 121$

compute student's t and test whether there is a significant difference in the two means.

Soln: -

$$s^2 = \frac{n_1 \sigma_1^2 + n_2 \sigma_2^2}{n_1 + n_2 - 2}$$

$$s^2 = \frac{(10 \times 100) + (10 \times 121)}{18} = 122.78$$

$$\therefore s = 11.0805$$

$$\text{we have } t = \frac{(\bar{x}_2 - \bar{x}_1)}{s \sqrt{1/n_1 + 1/n_2}}$$

$$t = \frac{60}{11.0805 \sqrt{0.1 + 0.1}} = 12.1081 \approx 12.11$$

This value of t is greater than the table value of t at all d.f of 18.

Thus the hypothesis that there is no significant in the two means is rejected at all significance levels.

3] A group of 10 boys fed on a diet A and another group of 8 boys fed on a diet B for a period of 6 months recorded the following increase in weights.

Diet A : 5 6 8 1 12 4 3 9 6 10

Diet B : 2 3 6 8 10 1 2 8

Test whether diet A and B differ significantly regarding their effect on increase in weight.

solⁿ: - Let the variable x correspond to the diet A and y to the diet B.

$$\bar{x} = \frac{\sum x}{n_1} = \frac{64}{10} = 6.4 \quad ; \quad \bar{y} = \frac{\sum y}{n_2} = \frac{40}{8} = 5$$

$$\sum_1^{n_1} (x - \bar{x})^2 = 102.4 \quad ; \quad \sum_1^{n_2} (y - \bar{y})^2 = 82$$

$$s^2 = \frac{1}{n_1 + n_2 - 2} \left\{ \sum_1^{n_1} (x - \bar{x})^2 + \sum_1^{n_2} (y - \bar{y})^2 \right\}$$

$$s^2 = \frac{1}{16} (102.4 + 82) = \frac{184.4}{16} = 11.525$$

$$\therefore s = 3.395$$

consider $t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

$$t = \frac{1.4}{3.395 \sqrt{\frac{1}{10} + \frac{1}{8}}} = 0.86935 \approx 0.87$$

But $t_{0.05}$ for 16 d.f. = 2.12 from the table; at 5% level of significance

$$\therefore t = 0.87 < 2.12$$

Thus we conclude that the two diets do not differ significantly regarding their effect on their increase in weight.

4] Two horses A and B were tested according to the time(s) to run a particular race with the following results.

Horse A: 28 30 32 33 33 29 34

Horse B: 29 30 30 24 27 29

Test whether you can discriminate between the two horses.

solⁿ: - let the variables x & y respectively correspond to horse A and horse B.

$$\bar{x} = \frac{\sum x}{n_1} = \frac{219}{7} = 31.3 \quad ; \quad \bar{y} = \frac{\sum y}{n_2} = \frac{169}{6} = 28.2$$

$$\sum_1^{n_1} (x - \bar{x})^2 = 31.43, \quad \sum_1^{n_2} (y - \bar{y})^2 = 26.84$$

$$s^2 = \frac{1}{n_1 + n_2 - 2} \left\{ \sum_1^{n_1} (x - \bar{x})^2 + \sum_1^{n_2} (y - \bar{y})^2 \right\}$$

$$s^2 = \frac{1}{11} (31.43 + 26.84) = 5.2973$$

$$\therefore s = 2.3016$$

consider $t = \frac{\bar{x} - \bar{y}}{s \sqrt{1/n_1 + 1/n_2}}$

$$t = \frac{(31.3 - 28.2)}{2.3016 \sqrt{1/7 + 1/6}} = 2.42$$

But $t_{0.05} = 2.2$ & $t = 2.42$

$$\therefore t = 2.42 > 2.2$$

Thus the discrimination between the horses is significant at 5% level.

5] A sample of 10 measurements of the diameter of a sphere gave a mean of 12cm and a standard deviation 0.15cm. Find the 95% confidence limits for the actual diameter.

solⁿ :- by data $n=10$, $\bar{x}=12$, $s=0.15$

also $t_{0.05}$ for 9 d.o.f = 2.262

confidence limits for the actual diameter is given by

$$\bar{x} \pm \left[\frac{s}{\sqrt{n}} \right] t_{0.05} = 12 \pm \frac{0.15}{\sqrt{10}} (2.262) = 12 \pm 0.1073$$

Thus 11.893 cm to 12.107 cm is the confidence limits for the actual diameter.

6] A sample of 11 rats from a central population had an average blood viscosity of 3.92 with a standard deviation of 0.61. On the basis of this sample, establish 95% fiducial limits for μ the mean blood viscosity of the central population ($t_{0.05} = 2.228$ for 10 d.o.f)

solⁿ: - by data $\bar{x} = 3.92$, $s = 0.61$, $n = 11$

$$\begin{aligned} 95\% \text{ fiducial limits for } \mu & \text{ are } \bar{x} \pm \left[\frac{s}{\sqrt{n}} \right] t_{0.05} \\ & = 3.92 \pm \frac{0.61}{\sqrt{11}} (2.228) \\ & = 3.92 \pm 0.41 = 3.51 \text{ and } 4.33 \end{aligned}$$

Thus 95% confidence limits for μ are 3.51 and 4.33.

Chi-Square distribution :-

Let O_i ($i=1, 2, 3, \dots, n$) and E_i ($i=1, 2, 3, \dots, n$) be the set of observed frequencies and expected frequencies respectively, then the Chi-Square distribution is defined as,

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

Note :- Calculated value of $\chi^2 <$ table values of χ^2 (at specified level of significance then the hypothesis is accepted otherwise rejected).

1. A die was thrown 60 times and the following frequency distribution was observed:

Faces	1	2	3	4	5	6
Frequency	15	6	4	7	11	17

Test whether the die is unbiased at 5% significance level. ($\chi^2_{0.05} = 11.07$).

→ The frequencies in the given data are the observed frequencies.

Assuming that die is unbiased, the expected no of frequencies for the no 1, 2, 3, 4, 5, 6 to appear on the face is $\frac{60}{6} = 10$ each

x	1	2	3	4	5	6
O_i	15	6	4	7	11	17
E_i	10	10	10	10	10	10

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{(15-10)^2}{10} + \frac{(6-10)^2}{10} + \frac{(4-10)^2}{10} + \frac{(7-10)^2}{10} + \frac{(11-10)^2}{10} + \frac{(17-10)^2}{10}$$

$$= \frac{(15-10)^2}{10} + \frac{(6-10)^2}{10} + \frac{(4-10)^2}{10} + \frac{(7-10)^2}{10} + \frac{(11-10)^2}{10} + \frac{(17-10)^2}{10}$$

$$\chi^2 = 13.6$$

2. A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured third class, 90 had secured second class and 20 had secured first class. Do these figures support the general examination result which is in the ratio 4:3:2:1 for the respective categories ($\chi^2_{0.05} = 7.8123$ to 3 degree of freedom).

→ The expected frequencies in the given data are

$$\frac{4}{10} \times 500, \frac{3}{10} \times 500, \frac{2}{10} \times 500, \frac{1}{10} \times 500$$

$$(or) 200, 150, 100, 50$$

x (Grade)	4	3	2	1
Observed frequency (O_i)	220	170	90	20
Expected frequency (E_i)	200	150	100	50

$$\chi^2 = \frac{\sum (O_i - E_i)^2}{E_i} = \frac{(220 - 200)^2}{200} + \frac{(170 - 150)^2}{150} + \frac{(90 - 100)^2}{100} + \frac{(20 - 50)^2}{50}$$

$$\chi^2 = 23.67 > 7.8123$$

Thus the hypothesis is rejected.

3. A Survey of 320 families with 5 children each revealed the following distribution.

No. of boys	5	4	3	2	1	0
No. of girls	0	1	2	3	4	5
No. of families	14	56	110	88	40	12

Is the result consistent with the hypothesis that male and female births are equally probable at 5% level of significance?

→ Given,

Number of families selected for the survey = 320
The probability of female and male birth is equal,

$$P = \frac{1}{2} = 0.5$$

$$q = 1 - P = 0.5$$

Number of children in the selected families, $n = 5$

No. of boys	5	4	3	2	1	0
No. of girls	0	1	2	3	4	5
No. of families	14	56	110	88	40	12

The statistical hypotheses is,

H_0 : The probability of female and male birth is equal.

H_1 : The probability of female and male birth is not equal.

Here Chi square distribution is used to test the hypotheses.

o.e. by the Binomial distribution

$$p(x) = {}^nC_x p^x q^{n-x}$$

$$p(x) = {}^5C_x (0.5)^x (0.5)^{5-x}$$

$$\Rightarrow p(x) = {}^5C_x (0.5)^5$$

The expected frequencies can be calculated for 320 families as

$$E(x) = 320 \times p(x) = 320 \times {}^5C_x (0.5)^5 = 10 {}^5C_x$$

$$E(0) = 10 \times {}^5C_0 = 10$$

$$E(1) = 10 \times {}^5C_1 = 50$$

$$E(2) = 10 \times {}^5C_2 = 100$$

$$E(3) = 10 \times {}^5C_3 = 100$$

$$E(4) = 10 \times {}^5C_4 = 50$$

$$E(5) = 10 \times {}^5C_5 = 10$$

No. of Boys	No. of Girls	Total observed Frequencies (O_i)	Expected Frequencies (E_i)	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
5	0	14	10	4	16	1.6
4	1	56	50	6	36	0.72
3	2	110	100	10	100	1
2	3	88	100	-12	144	1.44
1	4	40	50	-10	100	2
0	5	12	10	2	4	0.4

We have the Table value of χ^2 for 5 degrees of freedom at level of significance 5% from the Chi-Square table is 11.07.

$$\chi^2 = \sum \left[\frac{(O_i - E_i)^2}{E_i} \right]$$

$$\chi^2 = 7.16 < 11.02$$

Since the calculated χ^2 value is less than tabulated χ^2 value then the decision is fail to reject the H_0 [Accept H_0] that means both the male and female birth is equal.

4. 4 coins are tossed 100 times and the following results were obtained. Fit a binomial distribution for the data and test the goodness of fit ($\chi^2_{0.05} = 9.49$ for 4 degree of freedom)

No. of heads	0	1	2	3	4
Frequency	5	29	36	25	5

→ probability of getting heads $p = \frac{1}{2}$ $q = \frac{1}{2}$, $n = 4$
[∵ Binomial distribution]

$$p(x) = {}^nC_x p^x q^{n-x}$$

$$p(x) = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}$$

$$\because q = 1 - p$$

$$= 1 - \frac{1}{2}$$

$$\therefore p(x) = {}^4C_x \left(\frac{1}{2}\right)^4$$

Since 4 coins are tossed 100 times
[∵ $f(x) = N \times p(x)$]

$$f(x) = 100 \times p(x)$$

$$= 100 \times {}^4C_x \left(\frac{1}{2}\right)^4$$

$$f(x) = 6.25 \times {}^4C_x$$

put $x = 0, 1, 2, 3, 4$

No. of head	0	1	2	3	4
O_i	5	29	36	25	5
E_i	7	26	37	24	6

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(5-7)^2}{7} + \frac{(29-26)^2}{26} + \frac{(36-37)^2}{37} + \frac{(25-24)^2}{24} + \frac{(5-6)^2}{6}$$

$$= 1.15$$

$$\chi^2 = 1.15 < \chi^2_{0.05} = 9.49$$

∴ Thus the hypothesis that the fitness is good can be accepted.

5. Test the hypothesis that the data follow a binomial distribution. A set of five similar coins is tossed 320 times and the result is:

No. of heads	0	1	2	3	4	5
Frequency	6	27	72	112	71	32

(Given $\chi^2_{0.05} = 11.07$).

→ probability of getting heads $p = \frac{1}{2}$, $q = \frac{1}{2}$, $n = 5$

$$p(x) = {}^nC_x p^x q^{n-x}$$

$$= {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$$

$$= {}^5C_x \left(\frac{1}{2}\right)^5$$

Since 5 coins are tossed 320 times

$$f(x) = 320 \times p(x)$$

$$= 320 \times {}^5C_x \left(\frac{1}{2}\right)^5$$

$$f(x) = 10 \times {}^5C_x$$

put $x = 0, 1, 2, 3, 4, 5$

No. of head	0	1	2	3	4	5
O_i	6	27	72	112	71	32
E_i	10	50	100	100	50	10

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(6-10)^2}{10} + \frac{(27-50)^2}{50} + \frac{(72-100)^2}{100} + \frac{(112-100)^2}{100} + \frac{(71-50)^2}{50} + \frac{(32-10)^2}{10}$$

$$= 78.68$$

$$\chi^2 = 78.68 > \chi^2_{0.05} = 11.07$$

Thus the hypothesis is rejected

6. Fit a poisson distribution for the following data and test the goodness of fit (given that $\chi^2_{0.05} = 7.815$ for 3 degree of freedom).

x	0	1	2	3	4
f	122	60	15	2	1

Soln.

x	f	$f(x)$
0	122	0
1	60	60
2	15	30
3	2	6
4	1	4
	$\Sigma f = 200$	$\Sigma fx = 100$

$$M = \frac{\Sigma fx}{\Sigma f} = \frac{100}{200} = 0.5 = m$$

Poisson distribution

$$p(x) = \frac{m^x e^{-m}}{x!} = \frac{(0.5)^x e^{-0.5}}{x!}$$

$$\begin{aligned} f(x) &= 200 \cdot p(x) \\ &= 200 \cdot \frac{(0.5)^x e^{-0.5}}{x!} \\ &= 121.3 \frac{(0.5)^x}{x!} \end{aligned}$$

Put $x = 0, 1, 2, 3, 4$

No. of heads	0	1	2	3	4
O_i	122	60	15	2	1
E_i	121	61	15	3	0

$$\chi^2 = \frac{\Sigma (O_i - E_i)^2}{E_i} = \frac{1}{121} + \frac{1}{61} + 0 + 0 = 0.025$$

$$\chi^2 = 0.025 < 7.815$$

Thus the hypothesis that the fitness is good can be accepted.

Since the last expected frequency is 0 we shall club with previous one
 $O_i = 2 + 1 = 3$
 $E_i = 3 + 0 = 3$

7. The no of accidents per day (x) as recorded in a textile industry over a period of 400 days is given below. Test the goodness of fit in respect of poisson distribution of fit to be given data.
 $[\chi^2_{0.05} = 9.49$ for 4 degree of freedom]

x	0	1	2	3	4	5
f	173	168	37	18	3	1

→

x	f	fx
0	173	0
1	168	168
2	37	74
3	18	54
4	3	12
5	1	5
	$\Sigma f = 400$	$\Sigma fx = 313$

$$m = \frac{\Sigma fx}{\Sigma f} = \frac{313}{400} = 0.7825$$

Poisson distribution,

$$P(x) = \frac{m^x e^{-m}}{x!} = \frac{(0.7825)^x e^{-0.7825}}{x!} = \frac{183.36 (0.78)^x}{x!}$$

put $x = 0, 1, 2, 3, 4, 5$

x	0	1	2	3	4	5
O_i	173	168	37	18	3	1
E_i	183	143	56	15	3	0

$$3+1=4$$

$$3+0=3$$

last term having
 so $\frac{1}{0} = \infty$
 to avoid add terms

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(173-183)^2}{183} + \frac{(168-143)^2}{143} + \frac{(37-56)^2}{56} + \frac{(18-15)^2}{15} + \frac{(3-3)^2}{3} + \frac{(1-0)^2}{0}$$

$$\chi^2 = 12.3 > 9.49$$

Thus the hypothesis that the fitness is good is rejected